20261022 Quiz

for (MATH102) Calculus II

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Problem 3.46. Let $\mathbf{F}(x,y,z) = (xy^2, yz^2, zx^2)$. Find $\operatorname{curl}\mathbf{F}(1,2,3)$ and $\operatorname{div}\mathbf{F}(1,2,3)$.

Solution. Using the definitions of curl and divergence,

$$\begin{aligned}\nabla\times\mathbf{F} &= \left(\frac{\partial(zx^2)}{\partial y} - \frac{\partial(yz^2)}{\partial z}, \frac{\partial(xy^2)}{\partial z} - \frac{\partial(zx^2)}{\partial x}, \frac{\partial(yz^2)}{\partial x} - \frac{\partial(xy^2)}{\partial y}\right)\\ &= (-2yz, -2xz, -2xy),\\ \nabla\cdot\mathbf{F} &= \frac{\partial(xy^2)}{\partial x} + \frac{\partial(yz^2)}{\partial y} + \frac{\partial(zx^2)}{\partial z} = y^2+z^2+x^2.\end{aligned}$$

Consequently,

$$\boxed{\nabla\times\mathbf{F}(1,2,3) = (-12,-6,-4)}, \qquad \boxed{\nabla\cdot\mathbf{F}(1,2,3) = 14}.$$

□

Problem 3.48. Verify that

$$\operatorname{curl}(u(x,y), v(x,y), 0) = (0, 0, v_x - u_y).$$

Solution. Regard $(u(x,y), v(x,y), 0)$ as a vector field on $\mathbb{R}^3$ independent of z. Then

$$\begin{aligned}\nabla\times(u,v,0) &= \left(\frac{\partial 0}{\partial y} - \frac{\partial v}{\partial z}, \frac{\partial u}{\partial z} - \frac{\partial 0}{\partial x}, \frac{\partial v}{\partial x} - \frac{\partial u}{\partial y}\right)\\ &= \boxed{(0,0,v_x-u_y)},\end{aligned}$$

as required.

□

Problem 4.1. Compute the following partial derivatives.

(a) f_{xx} and f_{zz} for $(x,y,z) \neq (0,0,0)$ if $f(x,y,z) = (x^2+y^2+z^2)^{-1/2}$.
(b) f_{xyz} if $f(x,y,z) = xy + yz + zx$.
(c) $g_{xxy} - g_{xyx}$ if $g\colon \mathbb{R}^3 \to \mathbb{R}$ is three times continuously differentiable.
(d) h_{x_j} and $h_{x_jx_k}$ if $h(\mathbf{X}) = \mathbf{A}\cdot\mathbf{X}$, where $\mathbf{A} = (a_1,\dots,a_n)$ is constant and $\mathbf{X} = (x_1,\dots,x_n)$.

Solution.

(a) Write $r^2 = x^2+y^2+z^2$. For $r > 0$,

$$f_x = -xr^{-3}, \qquad f_z = -zr^{-3},$$

so the product and chain rules give

$$f_{xx} = -r^{-3} + 3x^2 r^{-5} = \boxed{\frac{3x^2 - r^2}{r^5}},$$

$$f_{zz} = -r^{-3} + 3z^2 r^{-5} = \boxed{\frac{3z^2 - r^2}{r^5}}.$$

(b) Since $f_x = y + z$, we have $f_{xy} = 1$ and hence $\boxed{f_{xyz} = 0}$.

(c) Theorem 4.1 applies to the C^3 function g. Both derivatives differentiate twice with respect to x and once with respect to y, so $g_{xxy} = g_{xyx}$ and $\boxed{g_{xxy} - g_{xyx} = 0}$.

(d) Since $h(\mathbf{X}) = \sum_{i=1}^{n} a_i x_i$,

$$\boxed{h_{x_j} = a_j}, \qquad \boxed{h_{x_j x_k} = 0} \quad (1 \le j, k \le n).$$

□

Problem 4.3. Let $f(x, y) = (x^2 + y^2)^{3/2}$. Find f_{xy}.

Solution. The first partial derivative is

$$f_x(x, y) = 3x\sqrt{x^2 + y^2},$$

including at the origin, where direct differentiation gives $f_x(0, 0) = 0$. For $(x, y) \neq (0, 0)$, the chain rule yields

$$f_{xy}(x, y) = \frac{3xy}{\sqrt{x^2 + y^2}}.$$

At the origin, the definition of the partial derivative gives

$$f_{xy}(0, 0) = \lim_{h \to 0} \frac{f_x(0, h) - f_x(0, 0)}{h} = 0.$$

Thus

$$\boxed{f_{xy}(x, y) = \begin{cases} \dfrac{3xy}{\sqrt{x^2 + y^2}}, & (x, y) \neq (0, 0), \\ 0, & (x, y) = (0, 0). \end{cases}}$$

□

Problem 4.13. Let

$$f(x, y) = (x - 1)^2 + 2(y - 2)^2 + (y - 2)^3 = 1 - 2x + x^2 + 4y - 4y^2 + y^3.$$

(a) Calculate ∇f and find the two points where $\nabla f = 0$.

(b) Calculate the Hessian matrix of f at the two points in part (a).

(c) Determine whether f has a local maximum, minimum, or saddle at the points you found.

Solution.

(a) Differentiating the expression in shifted coordinates gives

$$\nabla f(x, y) = \big(2(x - 1),\ 4(y - 2) + 3(y - 2)^2\big).$$

Thus $x = 1$ and $(y - 2)(3y - 2) = 0$, so

$$\boxed{\mathbf{P} = (1, 2), \qquad \mathbf{Q} = (1, \tfrac{2}{3})}.$$

(b) The Hessian is

$$\operatorname{Hess} f(x,y) = \begin{pmatrix} 2 & 0 \\ 0 & 4+6(y-2) \end{pmatrix}.$$

In particular,

$$\boxed{\operatorname{Hess} f(\mathbf{P}) = \begin{pmatrix} 2 & 0 \\ 0 & 4 \end{pmatrix}}, \qquad \boxed{\operatorname{Hess} f(\mathbf{Q}) = \begin{pmatrix} 2 & 0 \\ 0 & -4 \end{pmatrix}}.$$

(c) The first Hessian is positive definite, so $\mathbf{P}$ is a $\boxed{\text{strict local minimum}}$. The second is indefinite, so $\mathbf{Q}$ is a $\boxed{\text{saddle}}$.

□

Problem 4.15. Write each quadratic function in the form $\mathbf{X} \cdot \mathbf{SX}$ for some symmetric matrix $\mathbf{S}$.

(a) $3x_1^2 + 4x_1x_2 + x_2^2$.

(b) $-x_1^2 + 5x_1x_2 + 3x_2^2$.

Solution. For $\mathbf{X} = (x_1, x_2)$ and symmetric $\mathbf{S} = (s_{ij})$, the quadratic form is

$$\mathbf{X} \cdot \mathbf{SX} = s_{11}x_1^2 + 2s_{12}x_1x_2 + s_{22}x_2^2.$$

Therefore the coefficients give

(a) $\boxed{\mathbf{S} = \begin{pmatrix} 3 & 2 \\ 2 & 1 \end{pmatrix}}$.

(b) $\boxed{\mathbf{S} = \begin{pmatrix} -1 & \frac{5}{2} \\ \frac{5}{2} & 3 \end{pmatrix}}$.

□

Problem 4.16. Let $f(x,y) = x^2 + 2xy + y^3$. Determine all local extrema.

Solution. The critical points satisfy

$$f_x = 2x + 2y = 0, \qquad f_y = 2x + 3y^2 = 0.$$

Substituting $x = -y$ into the second equation gives $y(3y-2) = 0$. Hence the only critical points are

$$\mathbf{P} = (0,0), \qquad \mathbf{Q} = (-\tfrac{2}{3}, \tfrac{2}{3}).$$

The Hessian is

$$\operatorname{Hess} f(x,y) = \begin{pmatrix} 2 & 2 \\ 2 & 6y \end{pmatrix}, \qquad \det(\operatorname{Hess} f) = 12y - 4.$$

At $\mathbf{P}$, its determinant is $-4 < 0$, so $\mathbf{P}$ is a saddle. At $\mathbf{Q}$, the leading principal minors are $2 > 0$ and $4 > 0$, so $\mathbf{Q}$ is a strict local minimum. Moreover,

$$f(\mathbf{Q}) = \frac{4}{9} - \frac{8}{9} + \frac{8}{27} = -\frac{4}{27}.$$

Thus the only local extremum is

$$\boxed{\text{a strict local minimum at } (-\tfrac{2}{3}, \tfrac{2}{3}), \quad f = -\tfrac{4}{27}};$$

there is no local maximum.

□

Problem 4.17. Let $f(x, y) = -x^3 + x^2 + xy + 3y^2$. Show that $f(0, 0)$ is a local minimum.

Solution. We have

$$\nabla f(x, y) = (-3x^2 + 2x + y,\, x + 6y), \qquad \nabla f(0, 0) = \mathbf{0}.$$

The Hessian at the origin is

$$\operatorname{Hess} f(0, 0) = \begin{pmatrix} 2 & 1 \\ 1 & 6 \end{pmatrix}.$$

Its leading principal minors are $2 > 0$ and $12 - 1 = 11 > 0$, so the Hessian is positive definite. The second derivative test therefore shows that $\boxed{f(0, 0) = 0}$ is a strict local minimum.

□

Problem 4.20. Suppose a C^2 function $f(x, y)$ satisfies

$$f_{xx} + f_{yy} = 0.$$

(a) Suppose also that f_{xx} is not zero at any point. Show that f does not have any local maximum or minimum values.

(b) Suppose that at every point (x, y) at least one of f_{xx}, f_{xy}, or f_{yy} is not zero. Show that f does not have any local maximum or minimum values.

Solution. The given equation says $f_{yy} = -f_{xx}$, so

$$\operatorname{Hess} f = \begin{pmatrix} f_{xx} & f_{xy} \\ f_{xy} & -f_{xx} \end{pmatrix}, \qquad \det(\operatorname{Hess} f) = -f_{xx}^2 - f_{xy}^2.$$

(a) Since $f_{xx} \neq 0$, the determinant is negative everywhere. Thus the Hessian is indefinite at every point, ruling out both local maxima and local minima of a C^2 function.

(b) Because $f_{yy} = -f_{xx}$, the stated assumption ensures that f_{xx} and f_{xy} cannot both vanish. Hence again $\det(\operatorname{Hess} f) < 0$ everywhere. The Hessian is indefinite, so there are no local extrema.

□

Problem 4.21. Let $f(x, y) = x^2 + 2y^2$ and $g(x, y) = x^4 + 2y^2$. Show that f and g attain strict local minimums at the origin. Show that $\operatorname{Hess} f(0, 0)$ is positive definite but $\operatorname{Hess} g(0, 0)$ is not positive definite.

Solution. Both functions are positive away from the origin and vanish there: for $(x, y) \neq (0, 0)$,

$$f(x, y) = x^2 + 2y^2 > 0, \qquad g(x, y) = x^4 + 2y^2 > 0.$$

Thus both have strict (indeed, global) minima at $(0, 0)$. On the other hand,

$$\operatorname{Hess} f(0, 0) = \begin{pmatrix} 2 & 0 \\ 0 & 4 \end{pmatrix}, \qquad \operatorname{Hess} g(0, 0) = \begin{pmatrix} 0 & 0 \\ 0 & 4 \end{pmatrix}.$$

The first matrix is positive definite. The second is not, since its quadratic form is zero on every vector $(t, 0)$. This shows that a positive definite Hessian is sufficient, but not necessary, for a strict local minimum.

□

Problem 4.22. Find the point on the plane

$$z = x - 2y + 3$$

that is closest to the origin, by finding where the square of the distance between the origin and a point of the plane is a local minimum.

Solution. For a point $(x, y, x - 2y + 3)$ on the plane, the squared distance is

$$D(x, y) = x^2 + y^2 + (x - 2y + 3)^2.$$

The equations for a critical point are

$$\begin{aligned} D_x &= 2x + 2(x - 2y + 3) = 0, \\ D_y &= 2y - 4(x - 2y + 3) = 0. \end{aligned}$$

Equivalently, $2x - 2y + 3 = 0$ and $-2x + 5y - 6 = 0$, giving $x = -\frac{1}{2}$, $y = 1$, and $z = \frac{1}{2}$. Moreover,

$$\operatorname{Hess} D = \begin{pmatrix} 4 & -4 \\ -4 & 10 \end{pmatrix}$$

is positive definite because its leading principal minors are 4 and 24. Since D is a positive definite quadratic polynomial up to translation, this critical point is its unique global minimum. Thus the closest point is

$$\boxed{\left(-\frac{1}{2}, 1, \frac{1}{2}\right)}.$$

□

Problem 4.24. Let $f(\mathbf{X}) = \mathbf{C} \cdot \mathbf{X}$ be a linear function from $\mathbb{R}^n$ to $\mathbb{R}$, and let $\mathbf{A} \in \mathbb{R}^n$. Show that the first- and second-order Taylor approximations to f at $\mathbf{A}$, p_1 and p_2, are equal to f.

Solution. Since f is linear,

$$\nabla f(\mathbf{X}) = \mathbf{C}, \qquad \operatorname{Hess} f(\mathbf{X}) = \mathbf{0}.$$

For every increment $\mathbf{H}$,

$$\begin{aligned} p_1(\mathbf{A} + \mathbf{H}) &= f(\mathbf{A}) + \nabla f(\mathbf{A}) \cdot \mathbf{H} \\ &= \mathbf{C} \cdot \mathbf{A} + \mathbf{C} \cdot \mathbf{H} = f(\mathbf{A} + \mathbf{H}), \\ p_2(\mathbf{A} + \mathbf{H}) &= p_1(\mathbf{A} + \mathbf{H}) + \tfrac{1}{2}\mathbf{H}^{\mathsf{T}}(\operatorname{Hess} f(\mathbf{A}))\mathbf{H} = p_1(\mathbf{A} + \mathbf{H}). \end{aligned}$$

Thus $\boxed{p_1 = p_2 = f}$.

□

Problem 4.25. Let $f(x, y, z) = e^y \ln(1 + x) + \sin z$, defined for $x > -1$.

(a) Show that f has no local extrema.

(b) Find the second-order Taylor approximation $p_2(h_1, h_2, h_3)$ to f at $(0, 0, 0)$.

Solution.

(a) We have

$$f_x(x, y, z) = \frac{e^y}{1 + x} > 0 \qquad (x > -1).$$

Thus ∇f never vanishes on the open domain, so f has no local extrema.

(b) At the origin,

$$f(\mathbf{0}) = 0, \qquad \nabla f(\mathbf{0}) = (1, 0, 1),$$

and

$$\operatorname{Hess} f(\mathbf{0}) = \begin{pmatrix} -1 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}.$$

For $\mathbf{H} = (h_1, h_2, h_3)$, Taylor's formula gives

$$\begin{aligned} p_2(\mathbf{H}) &= f(\mathbf{0}) + \nabla f(\mathbf{0}) \cdot \mathbf{H} + \tfrac{1}{2}\mathbf{H}^{\mathsf{T}}(\operatorname{Hess} f(\mathbf{0}))\mathbf{H} \\ &= \boxed{h_1 + h_3 - \frac{1}{2}h_1^2 + h_1 h_2}. \end{aligned}$$

□

Problem 4.26. Let

$$f(x, y, z) = \frac{1}{1 - xyz}.$$

Find the first-order Taylor approximation to f at $\mathbf{A} = (\frac{1}{2}, \frac{1}{2}, \frac{1}{2})$.

Solution. The point satisfies $xyz = \frac{1}{8}$, so

$$f(\mathbf{A}) = \frac{1}{1 - \frac{1}{8}} = \frac{8}{7}.$$

By the chain rule,

$$\nabla f(x, y, z) = \frac{(yz, xz, xy)}{(1 - xyz)^2}, \qquad \nabla f(\mathbf{A}) = \left(\frac{16}{49}, \frac{16}{49}, \frac{16}{49}\right).$$

Consequently, for $\mathbf{H} = (h_1, h_2, h_3)$,

$$\boxed{p_1(\mathbf{A} + \mathbf{H}) = \frac{8}{7} + \frac{16}{49}(h_1 + h_2 + h_3)}.$$

□

Problem 4.27. Let $f(x, y, z) = x^2 + xy + \frac{1}{2}y^2 + 2yz + z^3$.

(a) Show that ∇f is the zero vector at $\mathbf{0}$ and at one other point $\mathbf{A}$.

(b) Use Theorem 4.10 to show that f has a local minimum at $\mathbf{A}$.

(c) Find points arbitrarily near $(0, 0, 0)$ where f is positive and where f is negative, to show that f has a saddle at $(0, 0, 0)$.

Solution.

(a) The critical-point equations are

$$\nabla f(x, y, z) = (2x + y,\ x + y + 2z,\ 2y + 3z^2) = \mathbf{0}.$$

The first two imply $y = -2x$ and $z = x/2$. Substituting into the third gives

$$-4x + \frac{3}{4}x^2 = 0 \quad \Longleftrightarrow \quad x = 0 \text{ or } x = \frac{16}{3}.$$

Therefore the two critical points are

$$\boxed{\mathbf{0} = (0, 0, 0), \qquad \mathbf{A} = (\tfrac{16}{3}, -\tfrac{32}{3}, \tfrac{8}{3})}.$$

(b) The Hessian is

$$\operatorname{Hess} f(x,y,z) = \begin{pmatrix} 2 & 1 & 0 \\ 1 & 1 & 2 \\ 0 & 2 & 6z \end{pmatrix}.$$

At $\mathbf{A}$, its leading principal minors are

$$2 > 0, \qquad \det \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix} = 1 > 0, \qquad \det(\operatorname{Hess} f(\mathbf{A})) = 8 > 0.$$

Thus Theorem 4.10 implies that $\operatorname{Hess} f(\mathbf{A})$ is positive definite. By the second derivative test, $\boxed{\mathbf{A} \text{ is a strict local minimum}}$.

(c) Along the z-axis, $f(0,0,z) = z^3$, which is positive for $z > 0$ and negative for $z < 0$. Both occur arbitrarily close to the origin, where $f(\mathbf{0}) = 0$. Hence $\boxed{\mathbf{0} \text{ is a saddle}}$.

□

Problem 4.29. Let $f(\mathbf{X}) = \|\mathbf{X}\|^{-1}$ for $\mathbf{X} \neq \mathbf{0}$ in $\mathbb{R}^3$, and let $\mathbf{A}$ be a nonzero vector in $\mathbb{R}^3$.

(a) Find all first- and second-order partial derivatives of f.

(b) Find the second-order Taylor approximation to f at $\mathbf{A}$.

Solution. Write $\mathbf{X} = (x_1, x_2, x_3)$, $r = \|\mathbf{X}\| > 0$, and let δ_{jk} be 1 if $j = k$ and 0 otherwise.

(a) Differentiating $f = r^{-1}$ yields, for $1 \le j, k \le 3$,

$$\boxed{f_{x_j}(\mathbf{X}) = -\frac{x_j}{r^3}}.$$

Differentiating once more gives

$$\begin{aligned} f_{x_j x_k}(\mathbf{X}) &= -\delta_{jk} r^{-3} + 3 x_j x_k r^{-5} \\ &= \boxed{\frac{3x_j x_k - r^2 \delta_{jk}}{r^5}}. \end{aligned}$$

Equivalently,

$$\nabla f(\mathbf{X}) = -\frac{\mathbf{X}}{r^3}, \qquad \operatorname{Hess} f(\mathbf{X}) = \frac{3\mathbf{X}\mathbf{X}^{\mathsf{T}} - r^2 \mathbf{I}_3}{r^5}.$$

(b) For $\mathbf{H} \in \mathbb{R}^3$ near $\mathbf{0}$, the second-order Taylor polynomial is

$$\begin{aligned} p_2(\mathbf{A} + \mathbf{H}) &= f(\mathbf{A}) + \nabla f(\mathbf{A}) \cdot \mathbf{H} + \tfrac{1}{2} \mathbf{H}^{\mathsf{T}} (\operatorname{Hess} f(\mathbf{A})) \mathbf{H} \\ &= \boxed{\frac{1}{\|\mathbf{A}\|} - \frac{\mathbf{A} \cdot \mathbf{H}}{\|\mathbf{A}\|^3} + \frac{3(\mathbf{A} \cdot \mathbf{H})^2 - \|\mathbf{A}\|^2 \|\mathbf{H}\|^2}{2\|\mathbf{A}\|^5}}. \end{aligned}$$

In particular, the linear term has a minus sign.

□

Problem 4.30. Use Theorem 4.10 to show that the following matrices are positive definite.

(a) $\mathbf{A} = \begin{pmatrix} 2 & -1 \\ -1 & 1 \end{pmatrix}$.

(b) $\mathbf{B} = \begin{pmatrix} 2 & -1 & 0 \\ -1 & 1 & 0 \\ 0 & 0 & 4 \end{pmatrix}$.

(c) $\mathbf{C} = \begin{pmatrix} 2 & -1 & 0 \\ -1 & 1 & k \\ 0 & k & 6 \end{pmatrix}$, where $k^2 < 3$.

Solution. Theorem 4.10 characterizes positive definiteness of a symmetric matrix by the positivity of all leading principal minors.

(a) For $\mathbf{A}$, these minors are $2 > 0$ and $\det(\mathbf{A}) = 2 - 1 = 1 > 0$. Thus $\boxed{\mathbf{A} \succ 0}$.

(b) For $\mathbf{B}$, they are $2 > 0$, $1 > 0$, and $\det(\mathbf{B}) = 4 > 0$. Thus $\boxed{\mathbf{B} \succ 0}$.

(c) For $\mathbf{C}$, the first two are $2 > 0$ and $1 > 0$, while

$$\det(\mathbf{C}) = 2(6 - k^2) - 6 = 2(3 - k^2) > 0$$

by the hypothesis $k^2 < 3$. Hence $\boxed{\mathbf{C} \succ 0}$.

□

Problem 4.33. Show using Theorem 4.11 that the maximum value of

$$Q(x, y) = 3x^2 + 2xy + 3y^2$$

on the unit circle $x^2 + y^2 = 1$ is 4.

Solution. The unit circle is compact, and its constraint $g(x, y) = x^2 + y^2$ has $\nabla g \neq \mathbf{0}$ on the circle. Thus a constrained maximum satisfies Theorem 4.11:

$$(6x + 2y, 2x + 6y) = \lambda(2x, 2y).$$

The resulting homogeneous linear system is

$$\begin{pmatrix} 3 - \lambda & 1 \\ 1 & 3 - \lambda \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}.$$

A nonzero solution requires $(3 - \lambda)^2 - 1 = 0$, so $\lambda = 2$ or 4. For $\lambda = 4$, $y = x$ and the unit circle gives $(x, y) = \pm(1/\sqrt{2}, 1/\sqrt{2})$, at which $Q = 4$. For $\lambda = 2$, $y = -x$ and $Q = 2$. Comparing the candidates proves

$$\boxed{\max_{x^2+y^2=1} Q(x, y) = 4},$$

attained at $\boxed{(x, y) = \pm(1/\sqrt{2}, 1/\sqrt{2})}$.

□

Problem 4.35. Use the Lagrange multiplier method to find the point of the hyperplane

$$\mathbf{C} \cdot \mathbf{X} = 0$$

that is closest to a given point $\mathbf{A} \in \mathbb{R}^n$.

Solution. Suppose first that $\mathbf{C} \neq \mathbf{0}$, as required for a hyperplane. Minimize the squared distance $f(\mathbf{X}) = \|\mathbf{X} - \mathbf{A}\|^2$ under $g(\mathbf{X}) = \mathbf{C} \cdot \mathbf{X} = 0$. The Lagrange multiplier equations give

$$2(\mathbf{X} - \mathbf{A}) = \lambda \mathbf{C}, \qquad \mathbf{C} \cdot \mathbf{X} = 0.$$

Thus $\mathbf{X} = \mathbf{A} + \frac{\lambda}{2}\mathbf{C}$, and the constraint yields

$$0 = \mathbf{C} \cdot \mathbf{A} + \frac{\lambda}{2}\|\mathbf{C}\|^2, \qquad \lambda = -\frac{2\mathbf{C} \cdot \mathbf{A}}{\|\mathbf{C}\|^2}.$$

Hence the nearest point, the orthogonal projection of $\mathbf{A}$ onto the hyperplane, is

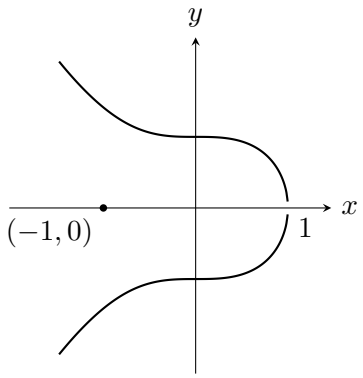
$$\boxed{\mathbf{X} = \mathbf{A} - \frac{\mathbf{C}\cdot\mathbf{A}}{\|\mathbf{C}\|^2}\mathbf{C}}.$$

Indeed, $\mathbf{A} - \mathbf{X}$ is normal to the hyperplane, so this is the unique global minimizer, and the distance is $\boxed{|\mathbf{C}\cdot\mathbf{A}|/\|\mathbf{C}\|}$. If $\mathbf{C} = \mathbf{0}$, the constraint is all of $\mathbb{R}^n$ and the nearest point is simply $\mathbf{A}$. □

Problem 4.38. Use the Lagrange multiplier method to find the points of the curve

$$x^3 + y^2 = 1$$

that are closest to the point $(-1, 0)$. See the sketch below.



Solution. Minimize the squared distance $D(x, y) = (x+1)^2 + y^2$ subject to $g(x, y) = x^3 + y^2 = 1$. Since $\nabla g = (3x^2, 2y)$ does not vanish on the curve, Theorem 4.11 applies. Its equations are

$$2(x+1) = 3\lambda x^2, \qquad 2y = 2\lambda y, \qquad x^3 + y^2 = 1.$$

If $y = 0$, then $x = 1$ and $D(1, 0) = 4$. If $y \neq 0$, then $\lambda = 1$, so

$$3x^2 - 2x - 2 = 0, \qquad x = \frac{1 \pm \sqrt{7}}{3}.$$

The positive root exceeds 1 and is incompatible with $y^2 = 1 - x^3 \geq 0$. Hence

$$t = \frac{1 - \sqrt{7}}{3}, \qquad y = \pm\sqrt{1 - t^3} = \pm\sqrt{\frac{5 + 10\sqrt{7}}{27}}.$$

To verify global minimality, the constraint requires $x \leq 1$, and on the curve

$$D(x, y) = (x+1)^2 + 1 - x^3 = x^2 + 2x + 2 - x^3.$$

This tends to ∞ as $x \to -\infty$, and its only critical point on $(-\infty, 1]$ is t (the other root exceeds 1). Since $D(t) < D(1) = 4$, these are the global minimizers. Thus the closest points are

$$\boxed{\left(\frac{1 - \sqrt{7}}{3},\ \pm\sqrt{\frac{5 + 10\sqrt{7}}{27}}\right)},$$

with minimum squared distance $\boxed{D_{\min} = (74 - 14\sqrt{7})/27}$.

□

Problem 4.39. Use the Lagrange multiplier method to find points of the level set

$$x^3 + y^2 + z^2 = 1$$

that are closest to the point $(-1, 0, 0)$.

Solution. Minimize $D(x, y, z) = (x + 1)^2 + y^2 + z^2$ subject to $g(x, y, z) = x^3 + y^2 + z^2 = 1$. The constraint gradient $(3x^2, 2y, 2z)$ never vanishes on the level set. The Lagrange equations are

$$2(x + 1) = 3\lambda x^2, \quad 2y = 2\lambda y, \quad 2z = 2\lambda z.$$

If $y = z = 0$, then $x = 1$ and $D = 4$. Otherwise $\lambda = 1$, so $3x^2 - 2x - 2 = 0$. As $y^2 + z^2 = 1 - x^3 \geq 0$, the only feasible root is

$$t = \frac{1 - \sqrt{7}}{3}.$$

The corresponding points satisfy

$$y^2 + z^2 = 1 - t^3 = \frac{5 + 10\sqrt{7}}{27}.$$

For any feasible point, $x \leq 1$ and $D = x^2 + 2x + 2 - x^3$. As in the direct one-variable minimization, this expression tends to ∞ as $x \to -\infty$ and has its minimum on $(-\infty, 1]$ at t, with $D(t) = (74 - 14\sqrt{7})/27 < 4$. Thus the complete set of closest points is the circle

$$\boxed{x = \frac{1 - \sqrt{7}}{3}, \qquad y^2 + z^2 = \frac{5 + 10\sqrt{7}}{27}}.$$

□

Problem 4.40. Let $f(x, y, z) = x^2 y^3 z^4$. Use the Lagrange multiplier method to find the maximum value of f on the plane

$$12x + 12y + 24z = 1.$$

Solution. As stated, the problem has *no maximum*: the plane is unbounded and f is unbounded above on it. Indeed, take $y = 1$ and $z = t$. The plane then requires $x = -\frac{11}{12} - 2t$, and

$$f\left(-\frac{11}{12} - 2t, 1, t\right) = \left(-\frac{11}{12} - 2t\right)^2 t^4 \longrightarrow \infty \qquad (t \to \infty).$$

Therefore the correct conclusion for the stated constraint is

$$\boxed{\text{No maximum exists on the plane.}}$$

If the intended problem also assumes $x, y, z \geq 0$, the feasible set is compact. On its boundary $f = 0$, while $f > 0$ in the interior. There the Lagrange equations can be written, after dividing by f, as

$$\left(\frac{2}{x}, \frac{3}{y}, \frac{4}{z}\right) = \mu(12, 12, 24).$$

Thus

$$x = \frac{1}{6\mu}, \qquad y = \frac{1}{4\mu}, \qquad z = \frac{1}{6\mu},$$

and substitution into $12x + 12y + 24z = 1$ gives $\mu = 9$. Under this *additional* condition, the maximum is attained at

$$\boxed{(x, y, z) = \left(\frac{1}{54}, \frac{1}{36}, \frac{1}{54}\right)}, \qquad \boxed{f_{\max} = \frac{1}{54^6\, 36^3}}.$$

This is not a maximum for the unrestricted plane in the original statement.

□